Apoptotic Effect of Crude Scorpion Venom on Head and Neck Squamous Cell Carcinoma Cell Line versus Normal Human Epithelial Cell Line (In-vitro Study)

Protocol submitted to

Faculty of Dentistry, Cairo University

for partial fulfillment of the requirements for the PhD Degree in in Oral and Maxillofacial Pathology

By

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I. Administrative information:

1. Title:

Apoptotic Effect of Crude Scorpion Venom on Head and Neck Squamous Cell Carcinoma Cell Line versus Normal Human Epithelial Cell Line (In-vitro Study)

2. Protocol registration

3. Protocol version:

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5. Roles and responsibilities:

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II. Introduction:

6a. Statement of the problem:

Squamous cell carcinoma arises from multiple anatomic subsites in the head and neck region. Current treatments involves a combination of surgery, radiotherapy and chemotherapy. Commonly used chemotherapeutic agents frequently affect normal human cells and cause severe side effects (Gerber, 2008 and Weeks et al., 2012). Moreover, cancer cells can develop resistance to these agents by cellular changes through multi-mechanism (Gatti & Zunino, 2005). Therefore, the development of new therapeutic agents that can be used for effective, specific and safe treatment has gained interest (Tong-ngam et al., 2015).

Scorpions and their venoms have been used in traditional medicine for thousands of years in China, India and Africa. The scorpion venom is a highly complex mixture of salts, nucleotides, biogenic amines, enzymes, mucoproteins, as well as peptides and proteins (e.g. neurotoxins). Yet, besides their neurotoxic activities, some have antitumor properties. In an increasing number of pre-clinical and experimental researches, cell cycle arrest and apoptosis induction, as well as inhibition of cancer proliferation and metastasis have been demonstrated to be caused by some purified peptides and proteins with the help of crude scorpion venom under in-vitro or in-vivo conditions (Ding et al., 2014).

PICO

P1**:** Cultured human HNSCC cell line.
P2**:** Cultured human epithelial cell line.
I: Application of crude scorpion venom on both cell lines.
C1: Cultured HNSCC cell line without venom application.
C2: Cultured normal epithelial cell line without venom application.
O: Assessment of cell viability, apoptosis and cell cycle phases in the 4 groups.

	Outcome name	**Measuring device**	**Measuring unit**
1ry outcome	Cell viability & cytotoxicity (MTT assay)	Microplate reader	%
2ry outcome	Expression of apoptotic marker (Puma)	ELISA plate reader	Ng/ml
3ry outcome	Analysis of phases of cell cycle (Propidium iodide staining)	Flow cytometer	KDa

Research Question:
Does crude scorpion venom has a selective apoptotic effect on head and neck squamous cell carcinoma cell line rather than normal epithelial cell line?

6b. Review of literature:

Squamous cell carcinoma arises from multiple anatomic subsites in the head and neck region. Head and neck squamous cell carcinoma (HNSCC) can arise from subsites within the oral cavity, oropharynx, hypopharynx, larynx, and nasopharynx. Worldwide, over 500,000 new cases of HNSCC are reported annually (Torre et al., 2015), with 40,000 new cases and 7890 deaths reported in the United States annually (Siegel et al., 2015). Nowadays, four standard methods are adopted for cancer treatment: surgery, radiotherapy, chemotherapy, and immunotherapy (Jemal et al., 2017).

Cancer cell migration and proliferation may be affected by the specific binding of some isolated peptides or proteins to the cancer cell membrane (Gao et al., 2008). Still, more potent anticancer drugs with fewer side effects are continuously searched for by oncologists since many of them have adverse effects on other organs, such as nervous system, heart, liver, bladder, lungs, and kidney (Moradi et al., 2018). The effects of new synthetic and natural substances in certain cancers, including products of animal, vegetable or mineral origin has been the focus of several researches in recent years. Among the animal substances being studied are components of some toxins such as those from scorpions (Ding et al., 2014) or bee (Oršolić, 2012).

The scorpion, which is one of the oldest creatures known, has existed on earth for more than 400 million years. Scorpions are known to be widely distributed all over the world, and there are over 1500 species that have been reported thus far (Simard & Watt, 1990). The Buthid scorpion Leiurus quinquestriatus (Egyptian scorpion) (Deathstalker scorpion) is found throughout Egypt, Israel, Syria, Jordan, and Saudi Arabia (Lucas & Meier, 1995). It is one of the commonest and most dangerous species in these areas. (Anderson, 1983).

In general, scorpion venom is considered to be a major health hazard. However, it has great potential if used in the correct way to fight various types of illness and is also being used by drug designing teams to synthesize an appropriate toxin-neutralizing vaccine (Ahn et al., 2000). In addition to this, scorpion venom and its derived molecules were found to have a high rate of success in inducing apoptosis in a wide variety of cancer cell lines. For example, the anticancer potential of scorpion venom has been reported against pancreatic and blood cancers (El-Ghlban et al., 2000 and Gupta et al., 2010).

Induced apoptosis and decrease in cell progression as well as disruption in normal architecture of the cancer cells are the characteristic features that were observed when these cells were treated with scorpion venom. Furthermore, the selective action of scorpion venom against cancer cells only, sparing normal cells, has also been reported (Díaz-García et al., 2013). However, a review article showed that the most of studies have focused on only inhibitory effects on cancer cells lines. Moradi et al., (2018) reported that the studies will be more valuable, if the inhibitory effects be assessed simultaneously in cancer cell line and normal cells.

Arpornsuwan et al., (2014) and Tong-ngam et al., (2015) studied BmKn-2 scorpion venom peptide and its antimicrobial and anticancer activities against human oral squamous cells carcinoma cell line (HSC-4). The BmKn-2 peptide killed HSC-4 cells through induction of apoptosis. Their results showed that scorpion venom and its derived peptides are promising chemotherapeutic agents for oral cancer treatment; which needed to be confirmed by further investigations. .

6c. Onjective:

The objective of this study is to investigate the apoptotic effect of crude scorpion venom on HNSCC cell line as well as normal human epithelial cell line, therefore providing a more selective, effective and less toxic possible treatment for such a malignancy.

III. Methods

A) Samples, intervention and outcomes

7. Calculated sample size

No sample size could be applied on cell line experiments as we generate millions of naturally randomized cells.

8. Description of study sample

1. Head and neck squamous cell carcinoma cell line.
2. Normal human epithelial cell line.
   Purchased form VACSERA, Cairo, Egypt and International Center for Advanced Research (ICTAR-Egypt) supplied with compatible nutrient media.

Inclusion Criteria: Only human cell lines.
Exclusion Criteria: Animal cell lines - Mixed tumors.

Sample preparation:

Cells will be sub-cultured in a 96-well plate.

1. Monolayer cells will be incubated for 48 hours at 37ºC in the 5% CO_2 incubator.
2. After 48 hours, cells will be fixed in wells for 1 hour in trichloro-acetic acid (50%-5 micro liter) which was added to the wells by a multichannel pipette.
3. Cells will be then washed with tap water and left to dry for 1 hour.
4. Sulphorhodamine B stain will be added (50 µl) and left for half an hour.
5. Excess stain will be washed with 5% glacial acetic acid and then attached stain was recovered with EDTA buffer (100 µl).
6. 50% of the samples will be treated with scorpion venom in complete growth media.

9. Intervention for each group

- Scorpion Venom: Crude scorpion venom powder of *Leiurus quinquestriatus* scorpion will be purchased from Egy venom company, Cairo, Egypt.
- HNSCC cell line and normal human epithelial cell line samples will be cultured with and without the addition of crude scorpion venom.

10. Outcomes

Assessment of cell viability (1ry outcome), apoptosis (2ry outcome) and cell cycle phases (3ry outcome) in the 4 groups.

Prioritization of Outcome	Outcome name	Measuring device	Measuring unit
Primary outcome	Cell viability and cytotoxicity (MTT assay)	Microplate reader	%
Secondary outcome	Expression of apoptotic marker (Puma)	ELISA plate reader	Ng/ml
Tertiary outcome	Analysis of phases of cell cycle (Propidium iodide staining)	Flow cytometer	KDa

a) MTT assay for cellular viability and cytotoxicity:

1. One ml of cells (50,000-100,000 cells/ml) will be plated into each well of 96-well culture plate for 24 hours before the MTT assay.
2. The cells will be incubated for 24 h in CO2 incubator.
3. After treatment of cells with antibiotics and antifungal for 24-72 h, experimental media will be removed, and cells will be washed with PBS.
4. Crude scorpion venom will be added for cell lines, incubated for 48 hours and then will be washed by PBS twice.
5. The cells will be incubated with medium containing 0.5 mg/ml MTT in CO2 incubator at 37^{o}C for 4 h.
6. The medium will be aspirated, and the formazan product will be solubilized with dimethyl sulfoxide (DMSO).
7. Absorbance at 570 nm will be measured for each well using a microplate reader (BioTek,Flx 800).
8. The percentages of cytotoxicity and cell viability are calculated using the following equations:
   % cytotoxicity = 1 - [means absorbance of treated cells/mean absorbance of negative control],
   % viability = 100 - % cytotoxicity.

b) Expression of apoptotic marker Puma:

Enzyme Linked Immunosorbent Assay (ELISA) will be used to evaluate the expression of the pro-apoptotic protein (Puma).

1. Cell lysis for cell culture samples will be carried by spinning down cells for 15 minutes at 1200 RPM.
2. Cell pellets are washed once in cold PBS. Cells are re-suspended in Lysis Buffer to a concentration of 1.5x106 cells/mL, are incubated for 1 hour at room temperature with gentle

shaking.

3. Cells are centrifuged at 200 x g for 15 minutes. The supernatant will be diluted at least 50-fold in 1X Assay Buffer (5 µL supernatant + 245 µL 1X Assay Buffer) for the assay.
4. Prepared biotin antibody is added to samples in different wells.
5. Then prepared Streptavidin horseradish peroxidase (HRP) solution is added after washing.
6. Tetramethylbenzidine (TMB) Solution is added to each well and incubated at room temperature.
7. Stop Solution is added to each well.
8. Readings will be done at 450 nm immediately.

C) Analysis of cell cycle phases with flow cytometry:

1. Crude scorpion venom will be added to flasks according to the pre-determined IC50 value and incubated for 48 hours.
2. Cells will be harvested by trypsnization and the single cell suspension will be prepared.
3. The cell suspension will be centrifuged for 15min at 1500 resolutions per minute to pellet cells.
4. 1 ml of cells were aliquoted in a 15 ml polypropylene 50µg/ml, V-bottom tube and 3 ml cold absolute ethanol were added.
5. Cells will be fixed and washed twice in PBS.
6. Annexin V and PI staining solution will be added to cell pellet and mixed.
7. RNase stock solution will be added to digest RNA and cells will be incubated.
8. The final cells will be filtered to remove aggregates prior to flow cytometry analysis.
9. Analysis will be performed using fluorescence-activated cell sorting (FASC) at a rate of 100/200 cells/sec. Data will be displayed as frequency histograms and dot plot.

B) Assignment to intervention

This study is non-randomized. Randomization is not applied on cell line experiments, as we generate millions of naturally randomized cells, where the amount of cells in each plate will be standardized and each of the 96-well plates will contain the same number of cells (about 25,000-30,000 cells/well).

11. Sequence generation

Not applied on cell line experiments.

12. Allocation concealment

Not applied on cell line experiments.

13. Implementation

Not applied on cell line experiments.

C) Blinding

14. Blinding

Not applied on cell line experiments.

D) Statistical methods

Statistical analysis will be performed with SPSS 16.0 (Statistical Package for Scientific Studies, SPSS, Inc., Chicago, IL, USA) for Windows. Numerical data will be described as mean and standard deviation or median and range. Categorical data will be described as number and percentage.

V-Statement of originality

Scorpion venoms are potential source of novel cancer therapeutic compounds. An increasing number of experimental and preclinical studies investigate the effect of crude scorpion venom and some purified proteins on a wide variety of human cancer cell lines. However, scarce studies investigate the effect of crude venom and specific purified peptide (BmKn2) on squamous cell carcinoma cell line. For the best of our knowledge, no studies investigate the apoptotic effect of venom on normal human epithelial cells. This will add to our knowledge about selectivity, effectiveness and toxicity of this possible treatment for such a malignancy.

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